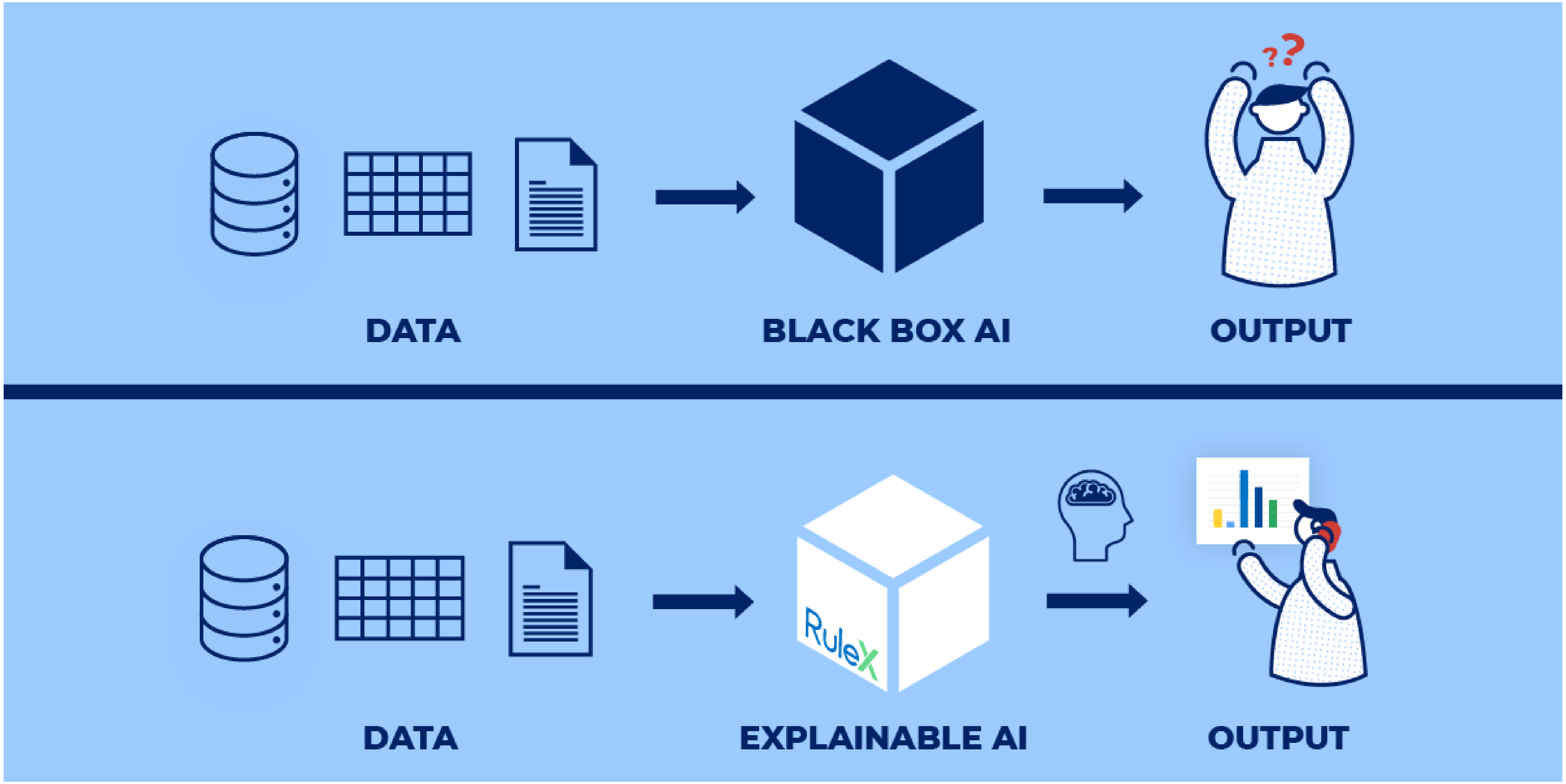
Explainable Machine Learning

Week 07


DATA
BLACK BOX AI
OUTPUT
RuleX
DATA
EXPLAINABLE AI
OUTPUT

Interpretable X Explainable ML

- **Interpretability** refers to the ability to observe cause-and-effect situations in a system, and, essentially, predict which changes will cause what type of shifts in the results (without necessarily understanding how it works).
- **Explainability** is basically the ability to understand and explain 'in human terms' what is happening with the model; how exactly it works under the hood.

FAT/ML: Fairness, Accountability, and Transparency in Machine Learning

Machine Learning raises novel challenges for ensuring non-discrimination, due process, and understandability in decision-making.

Policymakers, regulators, and advocates have expressed fears about the potentially discriminatory impact of Machine Learning.

Need further technical research into the dangers of inadvertently encoding bias into automated decisions.

There is increasing alarm that the complexity of machine learning may reduce the justification for consequential decisions to “the algorithm made me do it.”

Examples: Applications of explainable ML

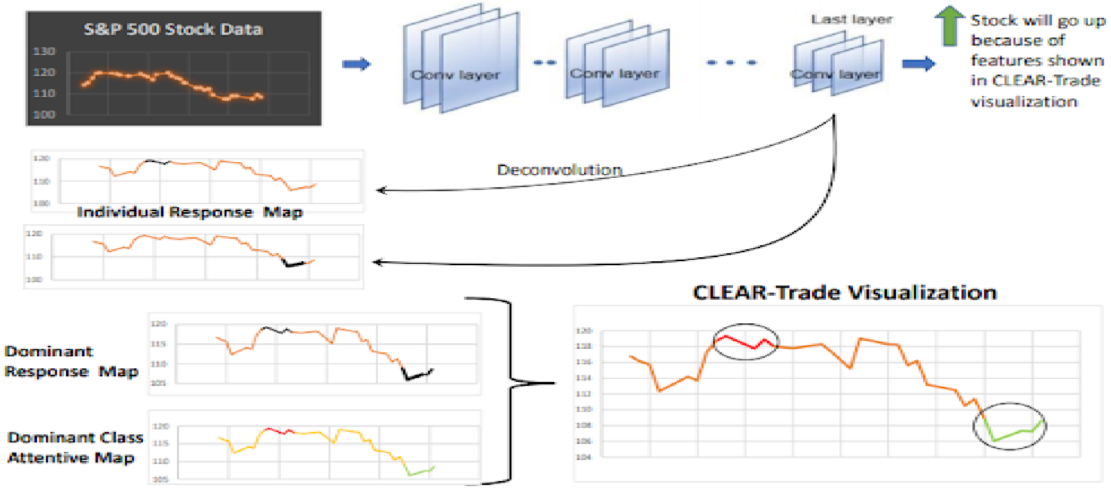
- When interacting with algorithmic decisions, users will expect and demand the same level of expressiveness from AI.
- A doctor diagnosing a patient may benefit from seeing cases that are very similar or very different.
- An applicant whose loan was denied will want to understand the main reasons for the rejection and what she can do to reverse the decision.
- A regulator will want to understand the behavior of the system as a whole to ensure that it complies with regulations.
- A developer may want to understand where the model is more or less confident as a means of improving its performance.

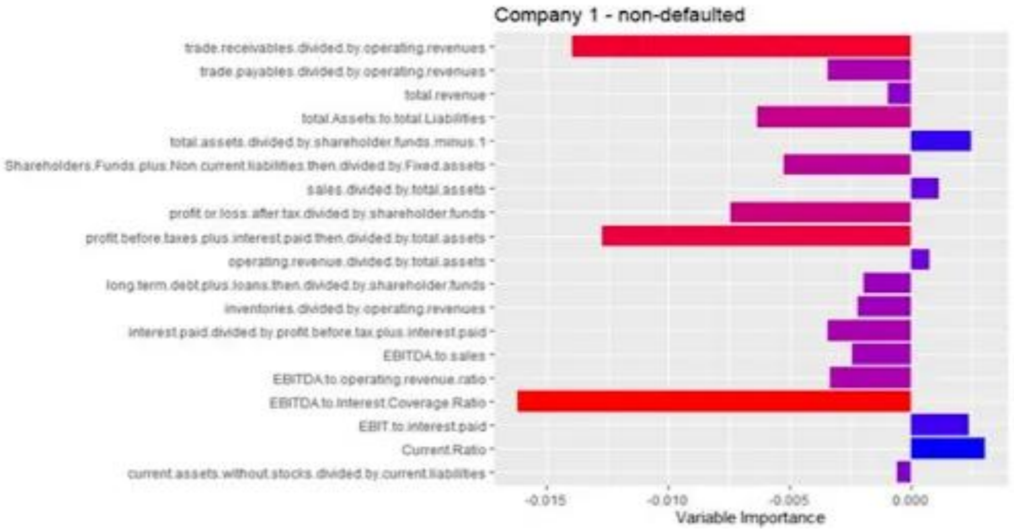
Finance

In finance, XAI is crucial for risk assessment, fraud detection, and explaining credit decisions. Here, regulatory compliance and customer trust play significant roles.

Trust building to convince investors and robustness against unfair outcomes are key goals. Both model logic and training data require examination.

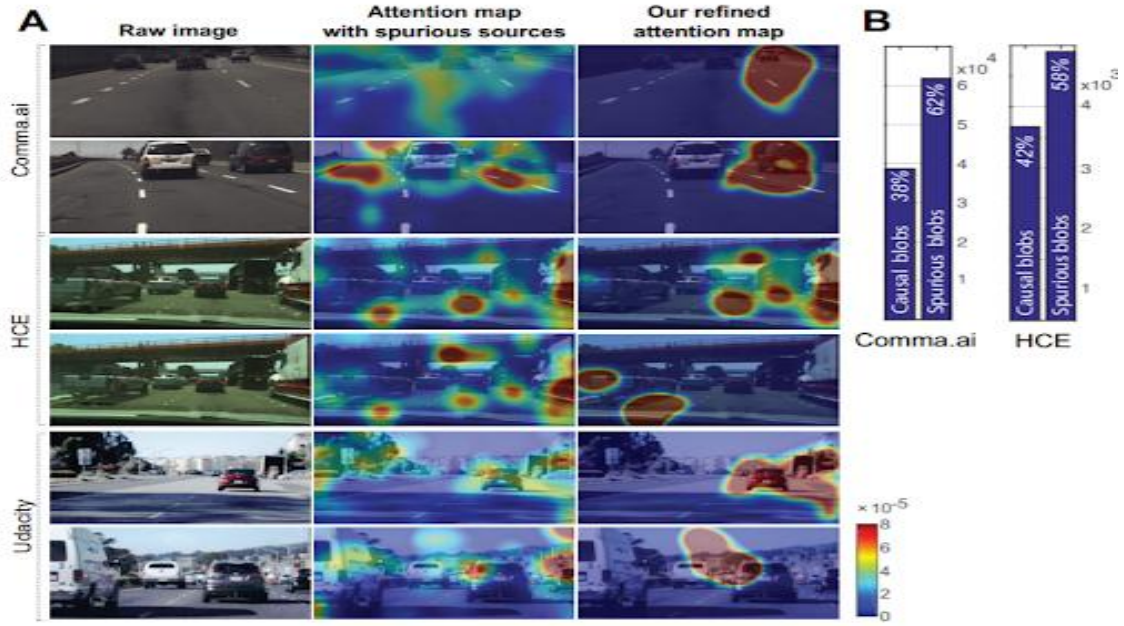
Explaining specific cases of credit default using prototypes shows promise. Heatmaps clarify stock market predictions but omit external drivers.

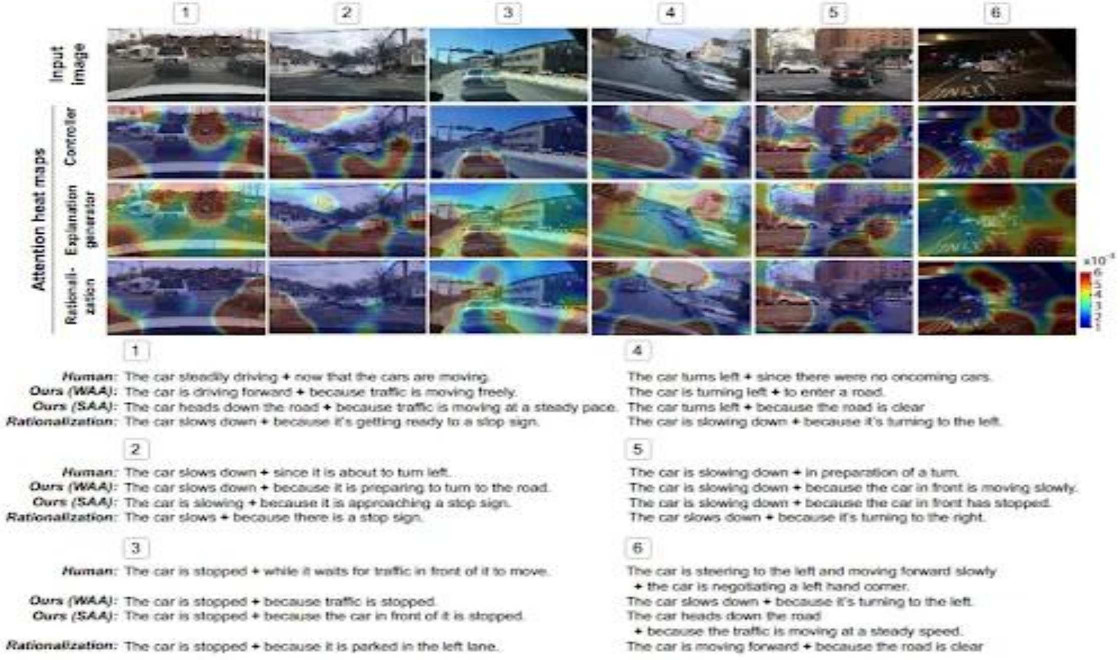




Autonomous Driving:

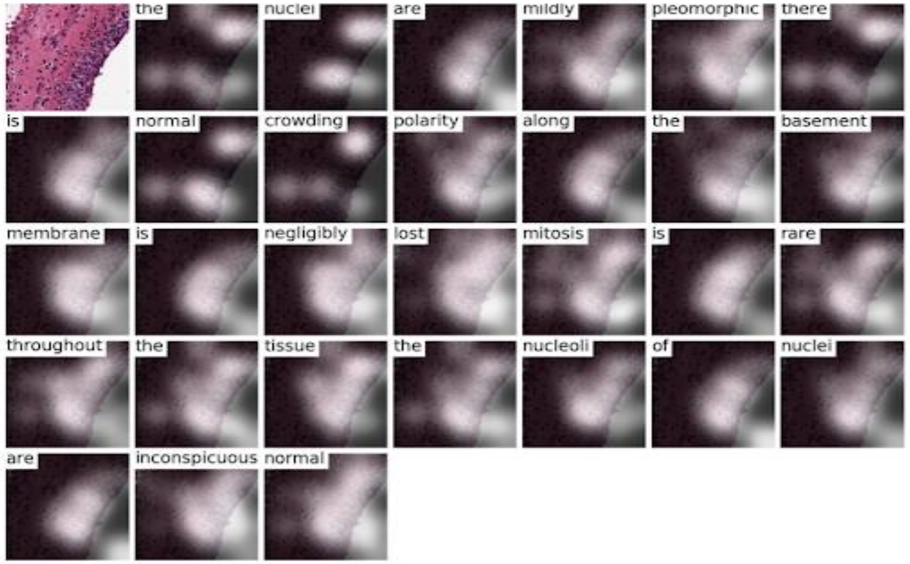
- Explainability in autonomous vehicles is vital for understanding vehicle decisions in real-time, which is crucial for safety and legal reasons.
- Explanations help operators intervene if needed and aid regulators in accountability assessments after accidents.
- Attention techniques identify failure points but lack reasoning details expected by engineers. Natural language augmentations show promise.

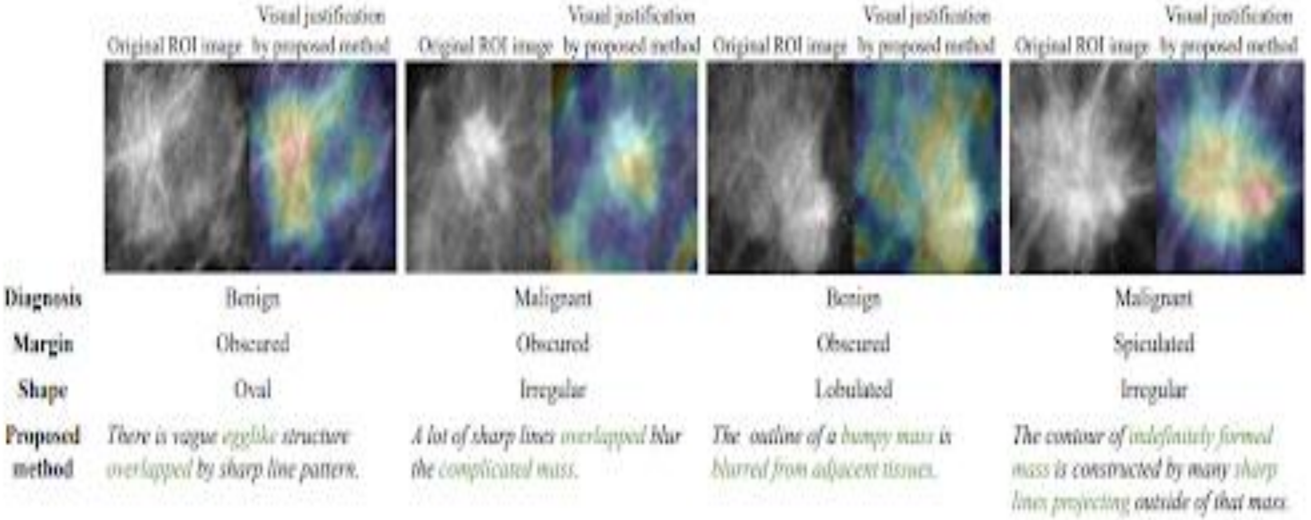




Healthcare

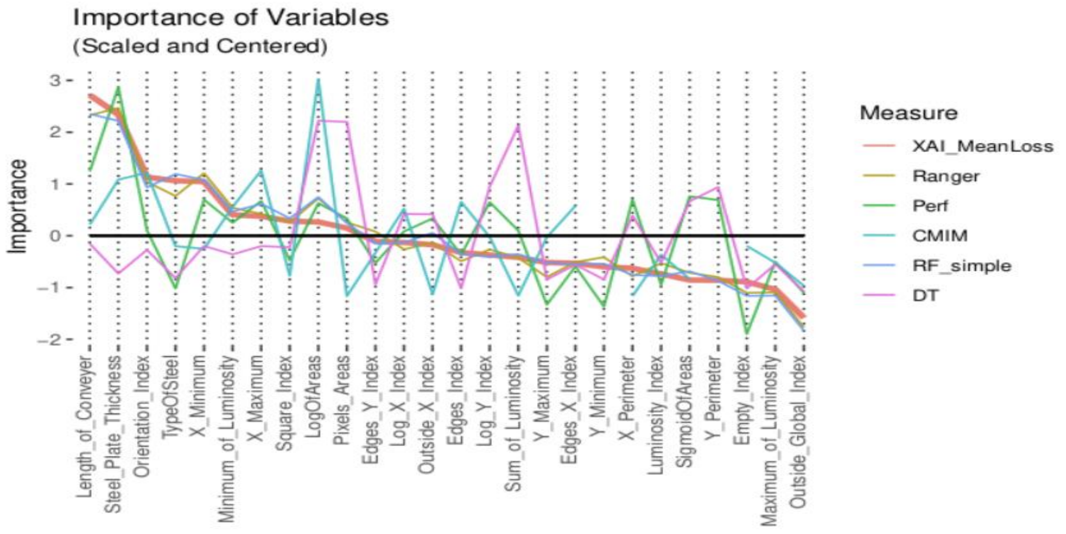
- In healthcare, XAI assists in diagnostic processes, treatment recommendations, and patient data analysis. The emphasis is on accuracy, reliability, and the ability to provide explanations that healthcare professionals can trust.
- Explanations help doctors calibrate trust and reconcile disagreements. Patient-facing education also matters. Domain constraints limit techniques.
- Pathology annotations clarify MRI scan classifier decisions for experts better than occlusion masks which have perceptual issues.





Manufacturing

- In the manufacturing sector, XAI is applied in predictive maintenance, quality control, and process optimization. Explanations are valuable for improving operational efficiency and decision-making.
- Explanations aid monitoring and redundancy rather than direct consumer value. Estimating production impact is the focal point.
- Ablative measures quantify feature contributions well but assume modularity



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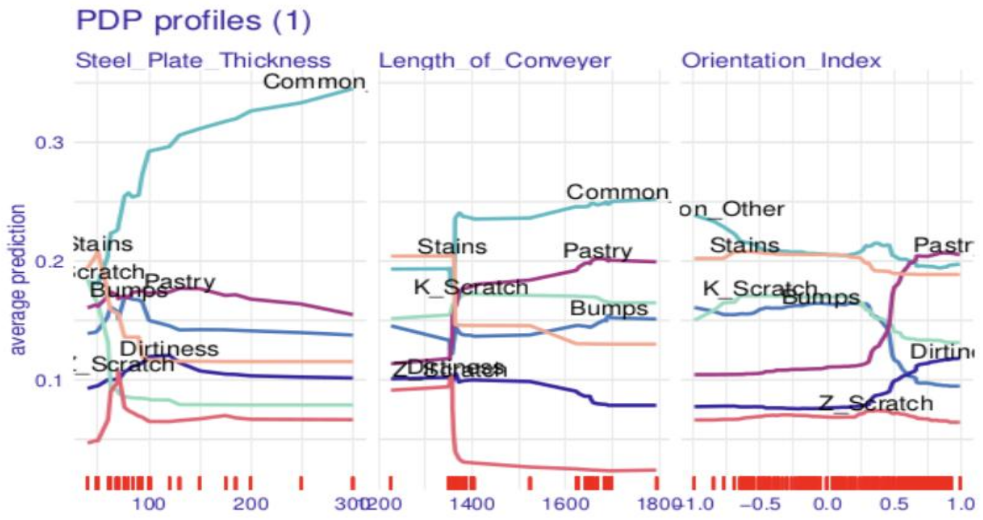


Why do we need interpretability?

Safety: system should provide sound decisions

Curiosity: understand something unexpected

Debugging: behaviour should be predictable

Optimality: optimize for true objectives

When we may not need interpretability

Low risk: no significant consequences

Awareness: problem is well-studied

Vulnerability: prevent people from gaming the system

Alternative 1: **Interpretable Models**

Use models that are intrinsically interpretable and known to be easy for humans to understand.

Examples: decision trees, decision rules and linear regression.

Alternative 2: **Interpreting Black Box Models**

Train a black box model and apply post-hoc interpretability techniques to provide explanations.

Focus: model-agnostic methods

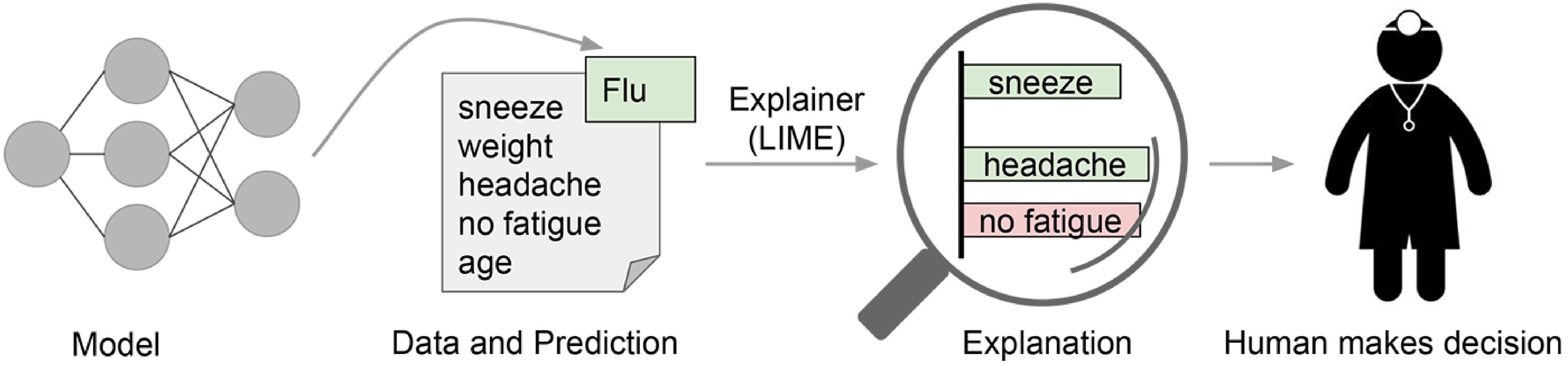
Examples: feature importance and accumulated local effects and explaining individual predictions with Shapley values and LIME.

LIME = Local Interpretable Model-agnostic Explanations

The LIME approach provides explanation for:

- an instance prediction of a model = the target
- in terms of input features = the drivers
- using importance scores = the explanation family
- computed through local perturbations of the model input = the estimator

Explaining individual predictions to a human decision-maker



Producing an Explanation - LIME Model for Images

Start with a normal image and use the black-box model to produce a probability distribution over the classes.

Then perturb the input in some way. For images, this could be hiding pixels by coloring them grey. Now run these through the black-box model to see the how the probabilities for the class it originally predicted changed.

Use an interpretable (usually linear) model like a decision tree on this dataset of perturbations and probabilities to extract the key features which explain the changes. The model is locally weighted — meaning that we care more about the perturbations that are most similar to the original image we were using.

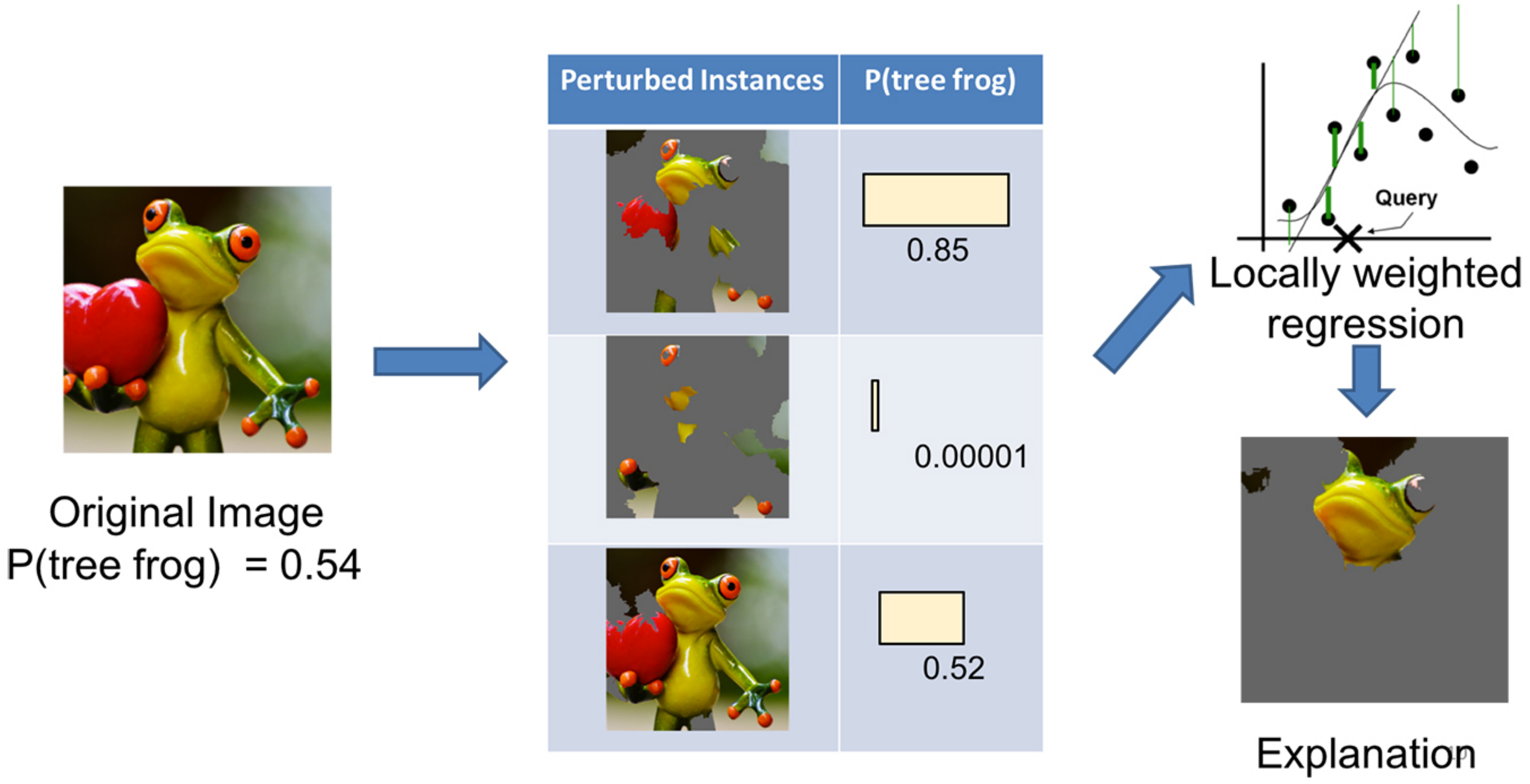
Output the features (in our case, pixels) with the greatest weights as our explanation.

Transforming an image into interpretable components

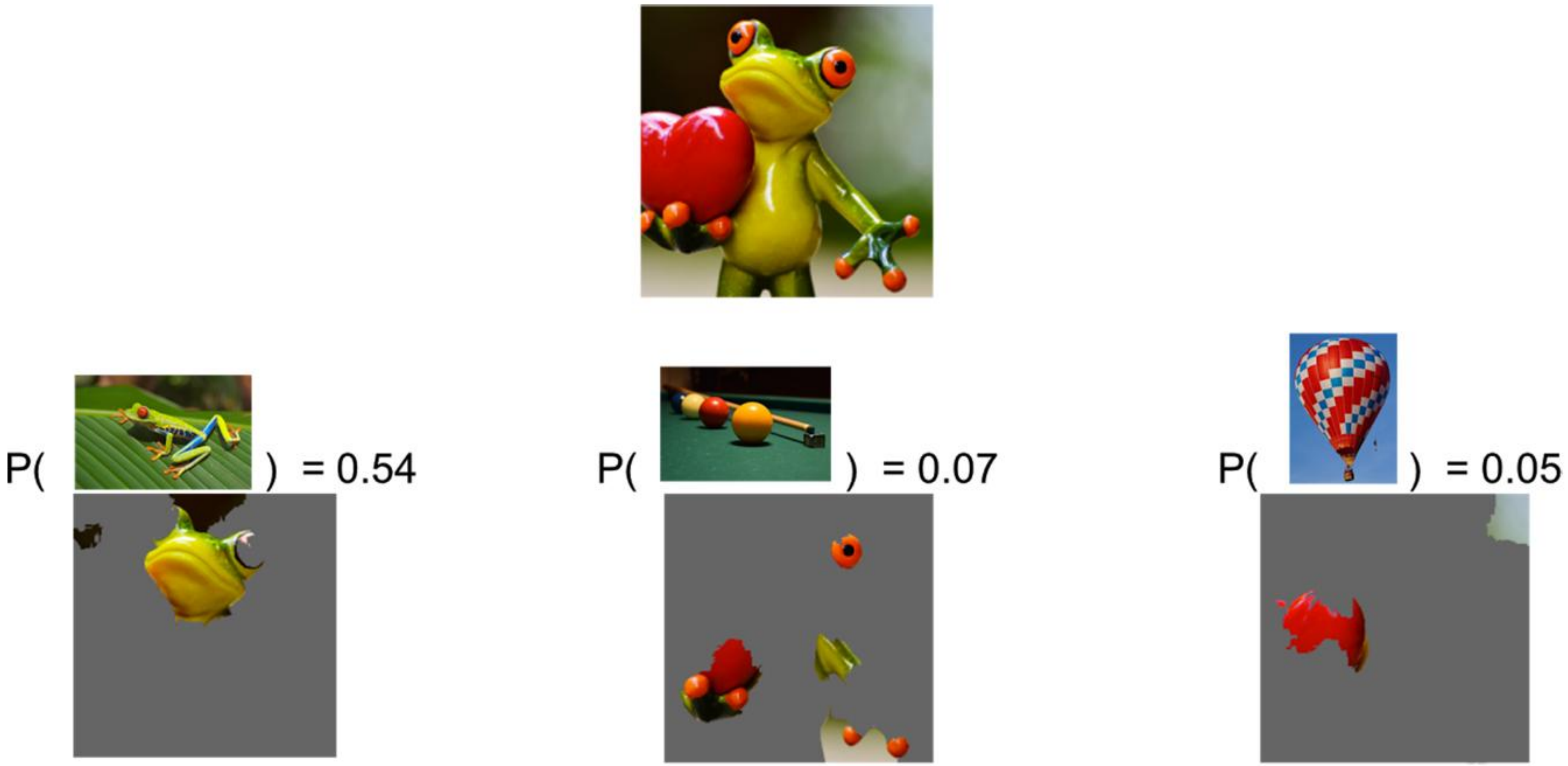
Original Image

Interpretable Components

Explaining a prediction with LIME



The top three predicted classes are "tree frog," "pool table," and "balloon"



Shapley Values

A prediction can be explained by assuming that each feature value of the instance is a “player” in a game where the prediction is the payout.

Shapley values – a method from coalitional game theory – tells us how to fairly distribute the “payout” among the features.

Shapley Values

The predicted price for a 50 m^2 2nd floor apartment with a nearby park and cat ban is €300,000.

Our goal is to explain how each of these feature values contributed to the prediction.

Definition: Shapley Value

The Shapley value is the average marginal contribution of a feature value across all possible coalitions.

Both the magnitude and the sign of the contributions are important:

- If a feature has a larger contribution than another, it has a larger influence on the model’s prediction for the observation of interest.
- The sign of the contribution indicates whether the feature contributes towards increasing (if positive) or decreasing (if negative) the model’s output.

SHAP = SHapley Additive exPlanations

The goal of SHAP is to explain the prediction of an instance x by computing the contribution of each feature to the prediction.

The SHAP explanation method computes Shapley values from coalitional game theory.

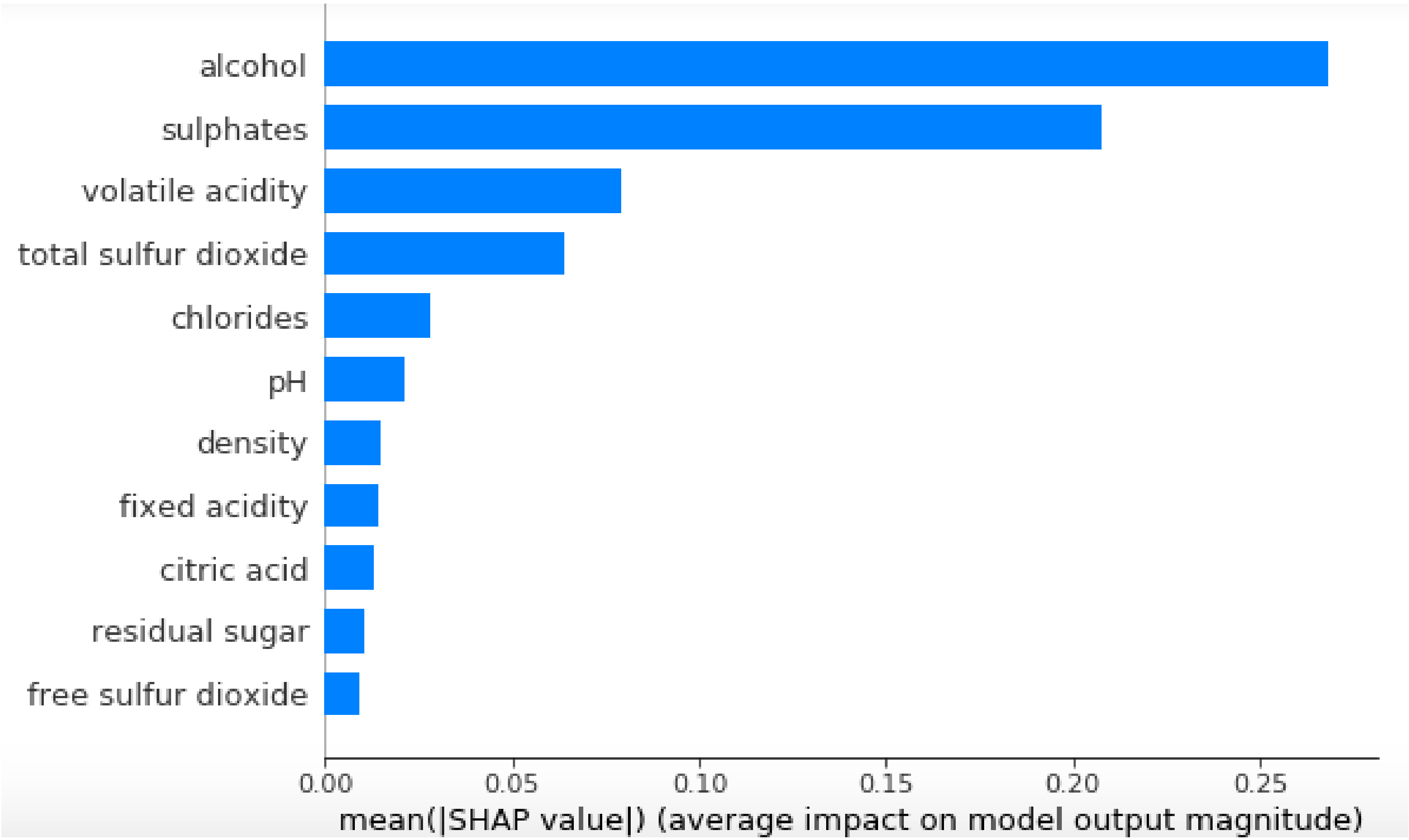
The feature values of a data instance act as players in a coalition. Shapley values tell us how to fairly distribute the "payout" (= the prediction) among the features.

A player can be an individual feature value, e.g. for tabular data. A player can also be a group of feature values.

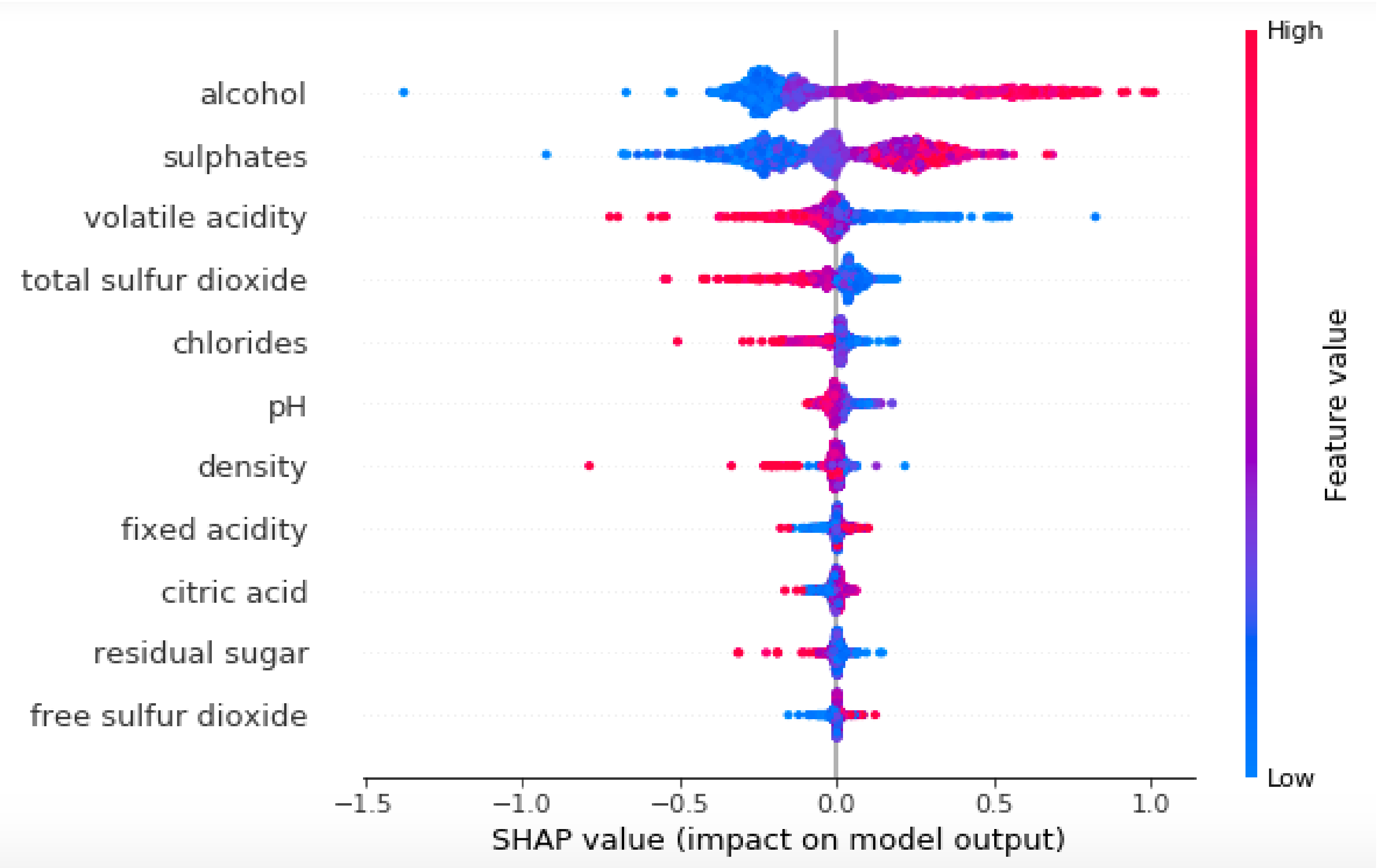
SHAP Code Example

```
import shap

from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor(max_depth=6, random_state=0, n_estimators=10)
model.fit(X_train, Y_train)

shap_values = shap.TreeExplainer(model).shap_values(X_train)
shap.summary_plot(shap_values, X_train, plot_type="bar")
```

Output: shap.summary_plot(shap_values, X_train, plot_type="bar")



Output: shap.summary_plot(shap_values, X_train)



ELI5

ELI5 is a Python library which allows to visualize and debug various Machine Learning models using unified API. It has built-in support for several ML frameworks and provides a way to explain black-box models.

Example: Who survived in the Titanic?

Features:

- **Age:** Age
- **Cabin:** Cabin
- **Embarked:** Port of Embarkation (C = Cherbourg; Q = Queenstown; S = Southampton)
- **Fare:** Passenger Fare
- **Name:** Name
- **Parch:** Number of Parents/Children Aboard
- **Pclass:** Passenger Class (1 = 1st; 2 = 2nd; 3 = 3rd)
- **Sex:** Sex
- **Sibsp:** Number of Siblings/Spouses Aboard
- **Survived:** Survival (0 = No; 1 = Yes)
- **Ticket:** Ticket Number

ELI5 Code Example

```
from eli5 import show_weights

clf = XGBClassifier()
clf.fit(train_xs, train_ys)
show_weights(clf, vec=vec)
```

Show_weights()
Output example

Weight	Feature
0.4278	Sex=female
0.1949	Pclass=3
0.0665	Embarked=S
0.0510	Pclass=2
0.0420	SibSp
0.0417	Cabin=
0.0385	Embarked=C
0.0358	Ticket=1601
0.0331	Age
0.0323	Fare
0.0220	Pclass=1
0.0143	Parch

Explaining Predictions

```
from eli5 import show_prediction

show_prediction(clf, valid_xs[1], vec=vec,
show_feature_values=True)
```

Show_prediction() Output Example

Contribution?	Feature	Value
+1.673	Sex=female	1.000
+0.479	Embarked=S	Missing
+0.070	Fare	7.879
-0.004	Cabin=	1.000
-0.006	Parch	0.000
-0.009	Pclass=2	Missing
-0.009	Ticket=1601	Missing
-0.012	Embarked=C	Missing
-0.071	SibSp	0.000
-0.073	Pclass=1	Missing
-0.147	Age	19.000
-0.528	<BIAS>	1.000
-1.100	Pclass=3	1.000

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